

Programme	B. Sc. Statistics				
Course Code	STA1FM105(1)				
Course Title	Quality Control				
Type of Course	MDC				
Semester	I				
Academic Level	100-199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	3	3	-	-	45
Pre-requisites	HSE level Mathematics Course				
Course Summary	To make students aware of Various Quality or standards in Industrial Production, Detecting, Controlling and Maintaining Quality and Total Quality Management				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Explain notion of Quality of products	U	C	Instructor-created exams / Quiz
CO2	Recall various meaning of Quality and critically evaluate ethical implications of statistical methods aligning with human values.	R	C	Quiz / Instructor-created exams
CO3	Explain causes of variation and Statistical Control and analyze data to help entrepreneurial decisions using critical thinking skills	U	F	Seminar Presentation / Group Tutorial Work/ Instructor-created exams
CO4	Construction of Control Charts and OC curves	Ap	C	Instructor-created exams / Home Assignments

CO5	Distinguish Process and Product Control	U	F	One Minute Reflection Writing assignments/ Instructor-created exams
CO6	Illustrate measures of central tendency using spread sheet.	Ap	P	Viva Voce/ Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C) # - Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)				

Module	Unit	Content	Hrs (36+9)	Marks (50)
I	Understanding Quality and Sources of Variation		9	15
	1	Meaning of Quality. Various Aspects of Quality.		
	2	Causes of Variation, assessing within and between sample		
	3	variation using Statistical Measures		
	4	Concept of Statistical Quality Control, Process Control and Product Control		
	Sections from References:			
II	Quantitative and Qualitative Variables		9	15
	5	Variables and Attributes.		
	6	Concept of Control Charts for Process Control		
	7	Structure of a Control Chart		
	8	Assessment of Statistical control using control charts		
	Sections from References:			
III	Construction of Charts		9	10
	9	Construction of $\bar{X}$ (mean) chart		
	10	Construction of R (Range) chart		
	11	Construction of σ (Standard Deviation) chart		
	12	Construction of P (Proportion Defective) chart		
	13	Construction of np (Number of Defectives) chart		
	14	Construction of C (Number of Defects) chart.		
	Sections from References:			
IV	Sampling Inspection Plan		9	10
	15	Acceptance Sampling Plan		
	16	Incoming Quality and Outgoing Quality		
	17	Acceptable Quality Level, Rejectable Quality Level, LTPD		
	18	AOQ, AOQL		
	19	Errors in Sampling Inspection Plan Producers and Consumers Risk		
	Sections from References:			
V	Open Ended Module: Spread sheet		9	

	1	Exercises to compute Arithmetic Mean, Range, Standard Deviation for a set of data, Basic concepts of Probability		
	Sections from References:			
Books and References:				
1. Statistical Quality Control Khanna Publishers				
2. Statistical Quality Control by Douglas C. Montgomery				
3. Fundamentals of Applied Statistics by Gupta S C and V. K. Kapoor				

Mapping of COs with PSOs and POs:

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	2	-	-	-	2	2	1	-	-	-	-
CO 2	2	-	-	-	-	3	2	2	-	-	-	3
CO 3	-	-	3	-	-	2	1	-			3	
CO 4	-	-	2	2	-	-	2		2	3		
CO 5	-	-	2	2	2	-	2	3	2	-	-	-
CO 6	3	2	-	-	-	2	3	-	-	3	-	-

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

18. Quiz / Assignment/ Quiz/ Discussion / Seminar
19. Midterm Exam
20. Programming Assignments (20%)
21. Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal	Assignm	Project	End Semester
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